

E4 Handoff Brief — The Type-1 Madness Grade: Extraction Rates of Rational Probing

Madness in the Probe sub-programme · banking run · issued by the lead instance, July 2026

You are the implementing instance for experiment E4 only. This brief is self-contained; do not request or assume other corpus documents. Implement fresh — do not consult, reconstruct, or reuse any prior sweep script.

1. Context (all you need)

The sub-programme studies **Madness**: structure or queries for which a probe cannot extract lawful answers at bounded cost. **Type 1 (M_1) is unbounded rationalisation cost**: objects that yield information forever, but at a floor-limited rate — resistance without terminus, not lawlessness.

The probe model: learning a real number x by rational approximation. The continued-fraction (CF) algorithm is the optimal rational prober (its convergents p_k/q_k are the best approximations, $|x - p_k/q_k| \asymp 1/(q_k q_{k+1})$), so $\ln q_k$ tracks the nats-about- x extracted after k questions. **The M_1 grade is the per-question information rate**:

$$\text{rate}(x) = \lim_k \frac{1}{k} \ln q_k, \quad \text{rate}_k(x) := \frac{1}{k} \ln q_k.$$

A lead preview found the classical structure lands cleanly on the grade; this run re-establishes it fresh and banks it. Stakes: the originator conjectured the golden ratio φ as maximal Type-1 Madness; the floor theorem makes that exact.

Convention pin (mandatory — bars below are indexed to it). CF $x = [a_0; a_1, a_2, \dots]$ with $a_0 = \lfloor x \rfloor$; denominators $q_0 = 1$, $q_1 = a_1$, $q_k = a_k q_{k-1} + q_{k-2}$. Internal check: for $\varphi = [1; 1, 1, \dots]$, q_4 **must equal 5** ($q_k = F_{k+1}$). Compute q_k in exact integer arithmetic from extracted CF terms; only term extraction uses arbitrary-precision floats.

2. The precision trap (mandatory guards)

A preview run fed 53-bit machine doubles into a high-precision CF: the CF of an exact dyadic rational terminates, and rounding near termination produced astronomical spurious quotients (rates ≈ 15 , i.e. $q_{60} \sim e^{900}$). Guards, all committed:

- **G1**. Random reals must carry entropy matching working precision: generate ≥ 800 random decimal digits per sample from a seeded generator. Never use float `random()` output as a real.

- **G2.** Working precision `mp.dps` 800 for $k_{\max} = 200$ (a typical real consumes ≈ 1.03 digits of precision per CF term).
 - **G3.** Round-trip: on a $\geq 10\%$ subsample, reconstruct x from its extracted terms and confirm agreement to $10^{-0.9\text{-dps}}$.
 - **G4.** Double-precision control: recompute a $\geq 10\%$ subsample at $2\times$ dps; rates must agree to 10^{-6} .
 - **G5.** Sanity: every extracted $a_i \geq 1$ for $i \geq 1$; for the known quadratics the periodic terms must match theory **exactly** (φ : all 1s; silver $1+\sqrt{2}$: all 2s; $\sqrt{2}$: $[1; 2, 2, 2, \dots]$); for $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$ the first 30 terms must match the pattern $a_{3m-1} = 2m$, others 1.
-

3. Pre-registered measures and bars

Copy this section into the script header before any check is written; report actuals against these fixed bars; **do not move bars** (standing ruling from E1). Checkpoints: $k \in \{30, 60, 120, 200\}$.

M1 — The universal floor (exact at finite k). Since $a_i \geq 1$ forces $q_k \geq F_{k+1}$, the floor holds at every k , not just in the limit: for **every** sampled or named x and every checkpoint, $\text{rate}_k(x) \geq \text{rate}_k(\varphi) - 10^{-12}$. Additionally φ 's own convergence: $|\text{rate}_{200}(\varphi) - 0.479594| \leq 10^{-4}$ (the k -corrected value: $\text{rate}_k(\varphi) \approx \ln \varphi + (\ln \varphi - \ln \sqrt{5})/k$, with $\ln \varphi = 0.4812118\dots$), and a fit of $\text{rate}_k(\varphi)$ against $1/k$ over $k \in [20, 200]$ recovering drift coefficient -0.3235 ± 0.01 .

M2 — Metallic fixed points. For $x_a = [a; a, a, \dots] = \frac{a+\sqrt{a^2+4}}{2}$: $\text{rate}(x_a) = \ln x_a$ — the number is learned at exactly its own log. Bars: $|\text{rate}_{200}(x_a) - \ln x_a| \leq 4 \times 10^{-3}$ for $a = 1..5$ (targets: 0.481212, 0.881374, 1.194763, 1.443635, 1.647231), with $|\text{rate}_k - \ln x_a|$ shrinking across checkpoints.

M3 — The tail control (the experiment's conceptual pin). $\sqrt{2} = [1; 2, 2, 2, \dots]$ shares the silver tail, so its rate must converge to $\ln(1+\sqrt{2}) = 0.881374$, **not** $\ln \sqrt{2} = 0.346574$: bars $|\text{rate}_{200}(\sqrt{2}) - \ln(1+\sqrt{2})| \leq 5 \times 10^{-3}$ and $|\text{rate}_{200}(\sqrt{2}) - \ln \sqrt{2}| \geq 0.5$. This pins the Reading that the grade reads the *probing process* (the tail), never the number's size. If $\sqrt{2}$ tracked $\ln \sqrt{2}$ the design itself would be wrong — report and stop, do not patch.

M4 — Typicality (Lévy). $n = 100$ random reals per G1, $k = 200$: sample mean of rate_{200} within 1.18657 ± 0.02 (Lévy's constant $\pi^2/(12 \ln 2)$); every sample obeys M1's floor; report the standard deviation and extremes.

M5 — The e anomaly. e 's partial quotients are unbounded (Euler), so its rate grows without bound — e is **anti-Mad**: more learnable per question than a typical real. Bars: $\text{rate}_k(e)$ strictly increasing across all four checkpoints; $\text{rate}_{200}(e) > 1.25$ (clear of Lévy). Report-only, no bar: a fit of $\text{rate}_k(e)$ against $\ln k$ over $k \in [50, 200]$, slope reported against the heuristic $1/3$. Companion, report-only: π (no known pattern; expect Lévy-ish).

M6 — Self-priced census (report-only, no bar, no grade). Count any sampled random real landing within 10^{-3} of its own $\ln x$ at $k = 200$ (i.e. $|\text{rate}_{200}(x) - \ln x| \leq 10^{-3}$). Feeds an open thread (“self-priced numbers”); strictly observational.

4. Prediction, grades on pass, escalation

Pre-registered prediction: all measures pass. This is a banking run of previewed classical structure. Any deviation is an implementation bug until proven otherwise; a **verified surviving** violation of the M1 floor or failure of the M3 tail control must not be patched — report it and return it to the register, since it would strike at the grade’s definition.

Grades on pass: floor, Lévy typicality, e -growth — **Fact** (classical: Hurwitz/Fibonacci bound, Lévy’s theorem, Euler’s CF of e ; carry these attributions). Metallic fixed-point identity — **Fact** (elementary). Tail control — **Fact**, pinning the process-not-value **Reading**. φ as the maximal-Madness pole of Type 1, and e as anti-Mad — **Reading**. M6 — report-only.

5. Protocol (carried from E1, with two additions)

1. Fresh implementation; PID-prefixed filename (`{PID}_e4_rates.py`); fixed recorded seed; pre-registered bars in the script header before checks are written; test-run to exit 0; report actuals against fixed bars.
 2. Deliverable: a results note `E4_Extraction_Rate_Results.md` mirroring E1’s format — status; reproducibility block (script, seed, versions, bars); per-measure results with actuals; a plain-language summary for Alex; an honesty-boundary paragraph (all asymptotics are classical; the grade M_1 , the tail-control design, and the Mad-pole/anti-Mad readings are the sub-programme’s; no leverage on any open problem). Do not edit the master document; do not stray into other experiments.
 3. **Non-vacuity clause (new, from E1’s addendum):** no check may compare a quantity to itself or restate a definition as a test; every committed bar must be capable of failing under an incorrect implementation. Audit your own checks against this before running.
 4. **Implementer-additions clause (new, from E1):** additions beyond the pre-registered set are welcome — label them explicitly as implementer additions, grade them separately, and never let them substitute for a committed measure.
-

Alexander Pisani & Claude (Anthropic) · Madness in the Probe sub-programme · July 2026.